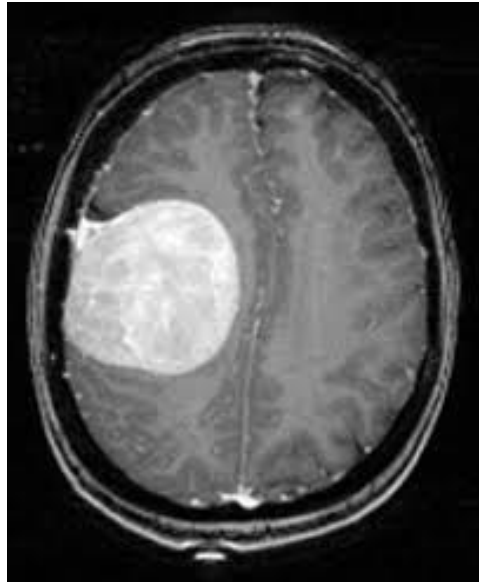
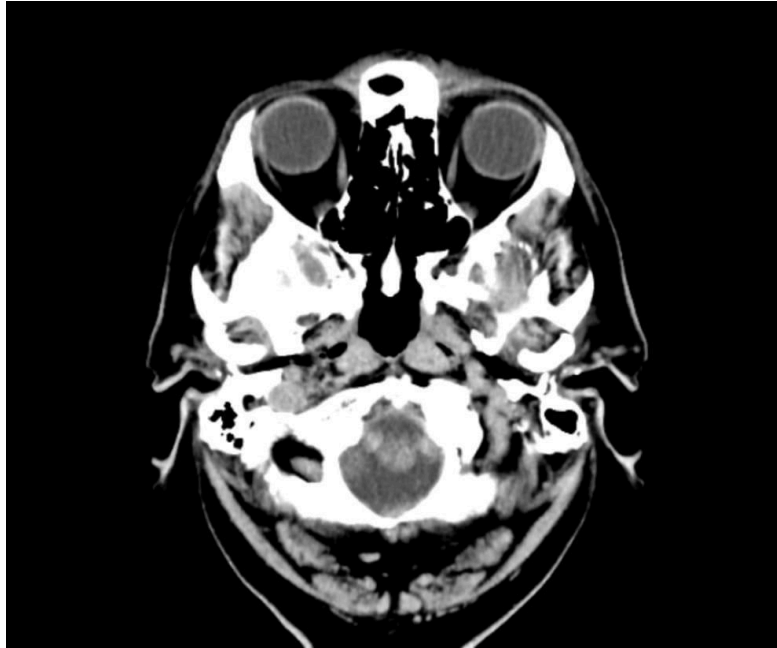


Lab 5

(Show input and output side by side for all problems)





1. **Apply** Fourier transform to transform any image (above) from the spatial domain to the frequency domain. Apply inverse Fourier transform to transform the image from the frequency domain to the spatial domain.

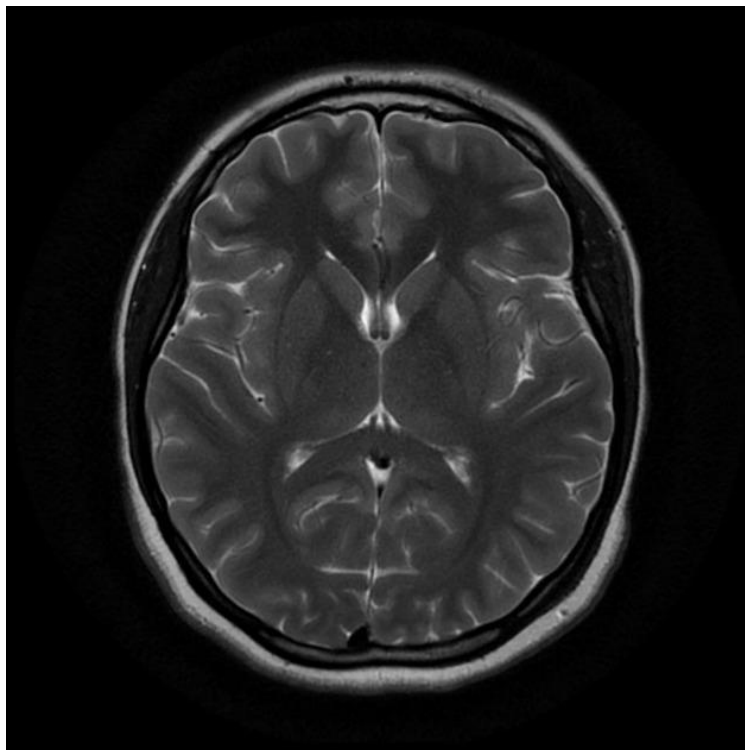


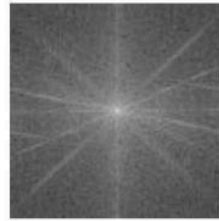
Figure 1: Brain MRI

2. **Apply** three types of high pass filtering in the frequency domain in **Figure 1** and find out which one is better to produce the enhanced image (sharpen) for the given image (output must show all steps as shown in **Figure 2**).
 - i. Ideal high pass filter (IHPF)
 - ii. Butterworth high pass filter (BHPF)
 - iii. Gaussian high pass filter (GHPF)
3. **Apply** three types of low pass filtering in the frequency domain in **Figure 1** and find out which one is better to produce the smoothen image for the given image (output must show all steps as shown in **Figure 2**).
 - i. Ideal low pass filter (ILPF)
 - ii. Butterworth low pass filter (BLPF)
 - iii. Gaussian low pass filter (GLPF)

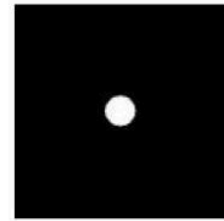
Ideal Low Pass Filter



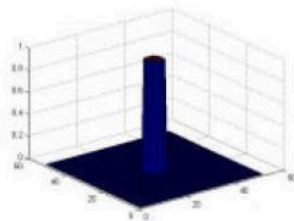
(a) Original Image



(b) Centered Fourier Spectrum



(c) Ideal Low Pass Filter



(d) Ideal Low Pass Filter



(e) Filtered Fourier Spectrum



(f) Processed Image

Figure 2: Sample Output of Ideal Low Pass Filter